Noise bends a polynomial fit

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Abstract

We measure how noise bends a polynomial fit, and find that its error grows as the square of the noise.

Introduction

State of the art

Polynomial fits are standard (Smith and Li 2020; Jones 2019; Kim 2017). Noise is known to bias them (Lee and Hall 2018). An earlier draft said so. A lost work claimed it first [to be resolved], and a late one agreed (Late 2024). A third said it again.

Results

How does noise change the fit?

The error grows with the noise

The error of the fit follows

$$e(\sigma) = 0.8\,\sigma^2 + 0.02 \tag{1}$$

and Eq. 1 holds at every noise level we ran, as Fig. 1 shows. Table 1 lists the errors. The fit follows Smith's method (Smith and Li 2020).

Table 1: The error of the fit at each noise level.

noise	rmse
0.1	0.03
0.2	0.05
0.4	0.15

Figure 1: The error of the fit grows with the noise (Lee and Hall 2018).

Discussion

Limits

The noise model is Gaussian, as Lee assumed (Lee and Hall 2018). A heavier tail would bend the fit further, and the next runs should try one.

Methods

Fitting

We fit by least squares, as Jones did (Jones 2019).

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Competing interests

The authors declare none.

Data availability

The data are synthetic and the project's runs made them.

Code availability

[TO BE GIVEN: the code repository, which setup has not settled]

References

Jones, Cy. 2019. "Least Squares for Curves." *Numerical Notes*, ahead of print. https://doi.org/10.5555/jones.

Kim, Fay. 2017. “A Survey of Fitting.” *Fitting Reviews*, ahead of print. https://doi.org/10.5555/kim.

Late, Ada. 2024. “A Late Agreement.” *Journal of Tests*, ahead of print. https://doi.org/10.5555/late.

Lee, Di, and Ed Hall. 2018. “Noise and Bias in Fits.” In *arXiv*. Preprint. https://arxiv.org/abs/1801.00001.

Smith, Ann, and Bo Li. 2020. “Polynomial Fits in Practice.” *Journal of Fits*, ahead of print. https://doi.org/10.5555/smith.